Prompt Strategies and Tool Selection for Automated Business Report Generation

1. Why a Simple Prompt May Fail

A simple prompt such as "Generate a business report from this data" often produces inconsistent results. The model may omit important sections, make assumptions about missing information, use an unsuitable format, or generate outputs that vary from one run to another. Business reports typically require a fixed structure, clear analysis, and reliable presentation, which a vague prompt may not enforce.

2. Improved Prompt Strategy

A structured or multi-step prompt is more effective. The prompt should clearly define the role, input data, required sections, output format, and evaluation criteria.

Example Multi-Step Prompt:
Step 1: Analyze the provided business data.
Step 2: Identify key metrics, trends, and anomalies.
Step 3: Create the following sections:
• Executive Summary
• Key Findings
• Data Analysis
• Risks and Challenges
• Recommendations
• Conclusion
Step 4: Present the report in a professional business format.
Step 5: Clearly distinguish facts from assumptions.

3. Tool Selection

Recommended Tool: ChatGPT (cloud-based LLM service)

4. Justification for Tool Selection

For business report generation, ChatGPT is typically the most practical choice because it provides strong language generation quality, supports complex prompts, requires minimal setup, and offers access to advanced models. It can quickly generate professional reports and is easier for business users to adopt compared to managing local infrastructure.

Alternative options:
• **Local Model (Ollama):** Better for privacy-sensitive data and offline deployment but may require more hardware resources and maintenance.
• **Google Colab:** Useful for experimentation, prototyping, and training workflows, but not ideal as a production deployment platform.

5. Trade-Off Analysis: Cloud vs Local Models

Factor	Cloud Models (e.g., ChatGPT)	Local Models (e.g., Ollama)
Performance	Usually higher-quality models	Depends on local hardware and model size
Setup	Minimal setup	Requires installation and configuration
Privacy	Data sent to external service	Data remains on local infrastructure
Cost	Subscription or API costs	Hardware and maintenance costs
Scalability	Easy to scale	Limited by local resources
Internet Requirement	Usually required	Can run offline